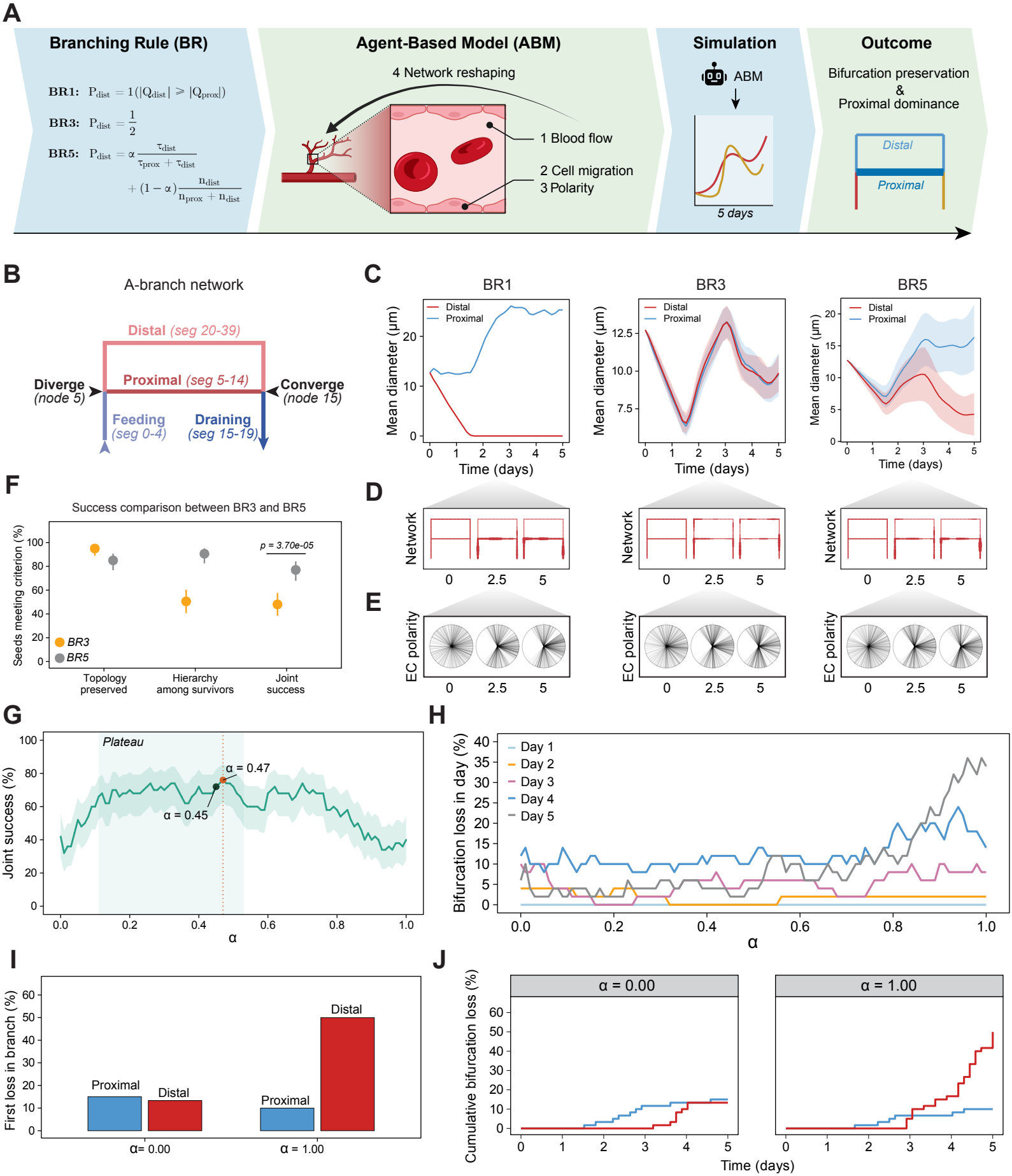
A
Branching Rule (BR)
BR1: $P_{dist} = 1(|Q_{dist}| \geqslant |Q_{prox}|)$
BR3: $P_{dist} = \frac{1}{2}$
BR5: $P_{dist} = \alpha \frac{\tau_{dist}}{\tau_{prox} + \tau_{dist}} + (1-\alpha)\frac{n_{dist}}{n_{prox} + n_{dist}}$
Agent-Based Model (ABM)
4 Network reshaping
1 Blood flow
2 Cell migration
3 Polarity
Simulation
ABM
5 days
Outcome
Bifurcation preservation
&
Proximal dominance
Distal
Proximal
B
A-branch network
Distal (seg 20-39)
Proximal (seg 5-14)
Diverge (node 5)
Converge (node 15)
Feeding (seg 0-4)
Draining (seg 15-19)
C
BR1
BR3
BR5
Distal
Proximal
Mean diameter (µm)
Time (days)
D
Network
E
EC polarity
F
Success comparison between BR3 and BR5
Seeds meeting criterion (%)
p = 3.70e-05
BR3
BR5
Topology preserved
Hierarchy among survivors
Joint success
G
Plateau
α = 0.47
α = 0.45
Joint success (%)
α
H
Day 1
Day 2
Day 3
Day 4
Day 5
Bifurcation loss in day (%)
α
I
First loss in branch (%)
Proximal
Distal
α= 0.00
α = 1.00
J
α = 0.00
α = 1.00
Cumulative bifurcation loss (%)
Time (days)